

Santhosh Narayanan Baburaman

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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026
B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024
Publication: Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

SKILLS

- **Languages & ML:** Python, SQL, R, PyTorch, Scikit-learn, XGBoost, Pandas, NLP, Transformer Models, Reinforcement Learning (Q-learning), Predictive Modeling, Anomaly Detection
- **Statistics:** Mixed-Effects Regression, Hypothesis Testing (FDR), Time Series (ARIMA, Prophet), statsmodels
- **Data & Cloud:** PostgreSQL, BigQuery, Snowflake, Redshift, GCP (Vertex AI, Kubeflow), Azure, LLM APIs, Git

EXPERIENCE

- Data Analyst Intern** | Philadelphia Department of Aviation Jun 2026 – Aug 2026
- Designed the department's first structured reporting and data-verification layer across 2 revenue streams covering ~30% of airport operating revenue, replacing ~60 hours/month of manual review
 - Engineered a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of billing data spanning 4 fiscal years, replacing manual Excel review and surfacing anomalies the prior process missed
 - Modeled a seasonal Z-score anomaly detector with peer-deviation and Benford's Law checks, and integrated ground transportation data with airline volumes to compute per-mode passenger capture rates
- IT Student Assistant** | USC Office of the Provost Jun 2025 – Present
- Provide IT support across 15+ administrative departments, configuring and reimaging devices for new users
- Founder & President** | ISE Analytics Student Association (IASA), USC Feb 2025 – May 2026
- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals
- AI Intern** | SmartInternz May 2023 – Jul 2023
- Led a 3-person team building a CNN-LSTM model that generates captions for 8,000 Flickr8k images in Python
- Azure Cloud Intern** | SmartKnower Sep 2022 – Oct 2022
- Deployed and managed virtual machines and containerized applications on Microsoft Azure cloud infrastructure

PROJECTS

- Healthcare Readmissions Prediction** (GCP Vertex AI, Kubeflow, Scikit-learn)
- Owned data preparation for a 30-day hospital readmission model on 8,038 patient records with a 17% positive class
 - Removed statistical outliers (z-score, modified z-score), engineered 3 clinical features, and applied SMOTE to the training split only; shipped preprocessing as versioned joblib artifacts
 - Deployed the selected gradient boosting model (AUPRC 0.86) through an 8-component Kubeflow pipeline on Vertex AI with model registry and endpoint serving
- Sleep & Cognition Research** | Student Researcher, Suen Lab, USC (Python, BigQuery) Sep 2026 – Present
- Reconstructed fragmented Fitbit sleep into continuous episodes (gap stitching within a night window) to build a second sleep dataset, and ran every analysis on both versions to test robustness
 - Linked 5 cognitive tests (Woodcock-Johnson battery, Stop-and-Go Switch, Figure Identification) to sleep across 7 time windows, using mixed-effects regression, between/within-person decomposition, FDR-corrected testing, and XGBoost GroupKFold predictive checks
- Automated Sales Analytics Copilot** (Python, PostgreSQL, Claude API, Streamlit) 🐙 GitHub
- Built an agentic LLM system with multi-step tool calling, text-to-SQL, and a 6-tool structured reasoning layer over a live PostgreSQL database
 - Developed ARIMA and Prophet forecasting with 80% confidence intervals and a 4-rule anomaly engine (z-score outliers, MoM trend reversals, margin compression)
 - Shipped a full-stack app on Streamlit with nightly GitHub Actions inference and automated Slack and email alerting
- eBay ML Challenge: Product Title NER** (Python, PyTorch, NLP) 🐙 GitHub
- Compared a BiLSTM-CRF against fine-tuned DistilBERT for Named Entity Recognition on 30K+ German eBay product titles, extracting brand, model, and color attributes; BiLSTM-CRF reached 0.78 macro-F1